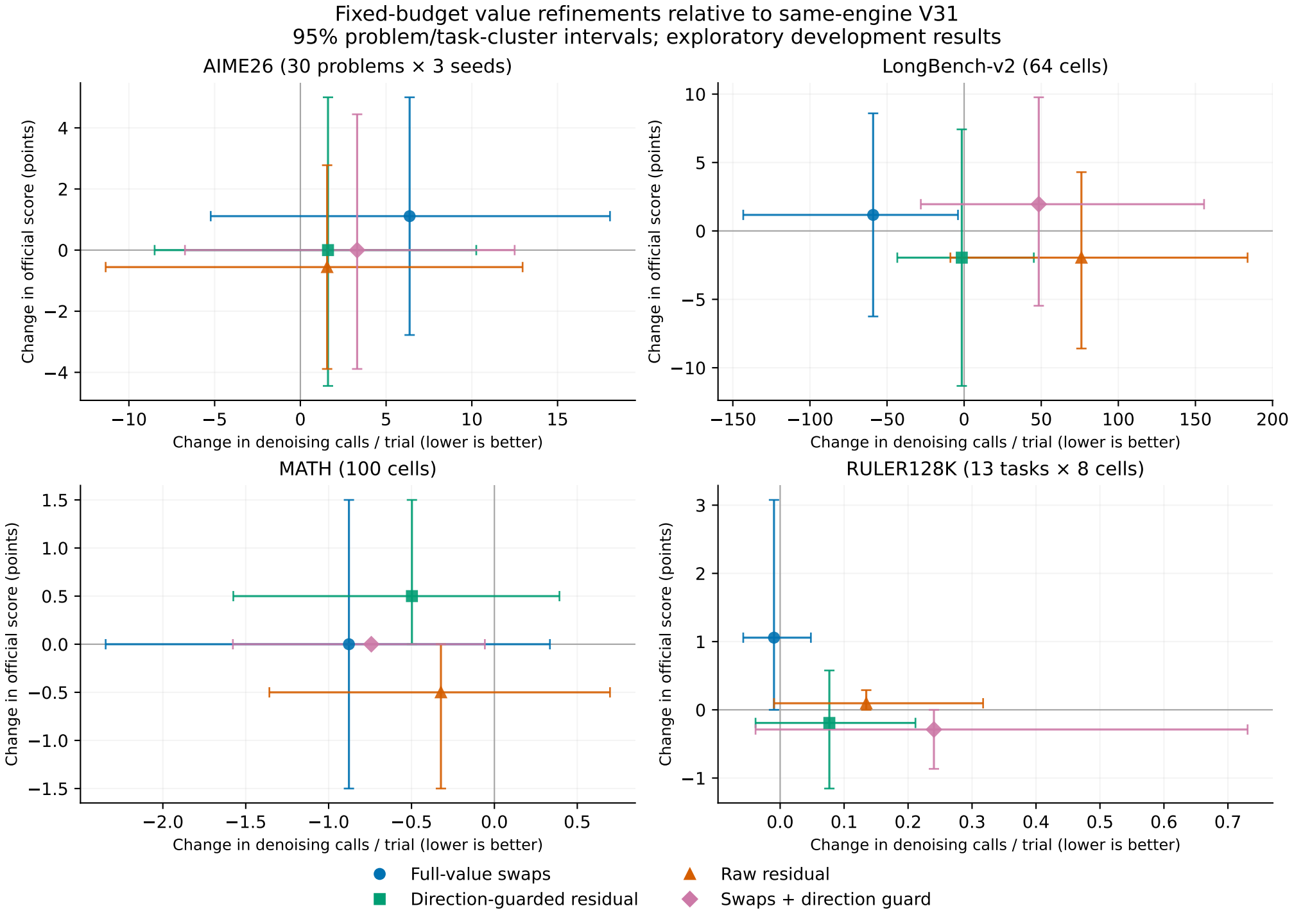
Fixed-budget value refinements relative to same-engine V31
95% problem/task-cluster intervals; exploratory development results
AIME26 (30 problems × 3 seeds)
LongBench-v2 (64 cells)
MATH (100 cells)
RULER128K (13 tasks × 8 cells)
Change in official score (points)
Change in denoising calls / trial (lower is better)
Full-value swaps
Direction-guarded residual
Raw residual
Swaps + direction guard